Arista Access Node WFD-ACC-31 Routing Protocol Flap Resolution SOP (Whitefield POP)

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5434

Description

The failure domain of routing protocol flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Arista Access Node WFD-ACC-31 at the Whitefield POP is attributed to the intricate interplay between the control and data planes of the Cisco ASR9000 Series platform. This complex interplay can lead to routing protocol instabilities, which, in turn, cause routing protocol flaps. The root cause of this issue can be attributed to the high-availability (HA) clustering configuration, which is designed to provide redundancy and ensure continuous network operation. However, in certain scenarios, this HA configuration can lead to unintended routing protocol interactions, resulting in routing protocol flaps. Furthermore, the use of advanced routing protocols such as BGP, OSPF, and ISIS, which are designed to provide scalable and flexible routing, can also contribute to the complexity of the issue. Additionally, the fact that the Arista Access Node WFD-ACC-31 is configured as a high-availability (HA) cluster, with multiple nodes participating in the cluster, can also increase the likelihood of routing protocol flaps. In summary, the failure domain of routing protocol flap on the Arista Access Node WFD-ACC-31 at the Whitefield POP is a complex issue, requiring a deep understanding of the underlying protocol and hardware mechanics.

Pre-requisite Checks

- Verify that the Whitefield POP has been placed in a 'maintenance mode' to prevent any further network traffic

- Check the status of redundant links to ensure they are online and functioning properly

- Notify the NOC team of the maintenance activity and obtain their approval

- Perform a backup of the node configuration to ensure that any changes made during the troubleshooting process can be easily rolled back

Potential Root Causes

Cause	Probability
BGP route churn due to high-availability (HA) clustering configuration	High
OSPF adjacency loss due to routing protocol instability	Medium
ISIS LSP flooding due to misconfigured routing protocols	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 123
Router ID is 10.10.10.10 BGP routing-table version is 123 Router ID
10.10.10.10, RIBs: 2 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ
```

```
Up/Down State PrefRcvd, Description: This command displays a summary
of the BGP routing table, including the number of routes received and
sent, as well as the number of BGP neighbors

Command: show ospf neighbor, Expected Output: OSPF neighbor table
Neighbor State Address Interface Pri DeadTime Time Up/Down
---------------------- 10.10.10.10 FULL/DR 10.10.10.10 Eth1/1 1
00:00:03 00:00:01 00:05:30 00:00:02, Description: This command
displays a list of OSPF neighbors, including their state and the
interface they are connected to

Command: show isis database, Expected Output: ISIS database LSA type:
1 LSA ID: IS-IS:100 LSA type: 2 LSA ID: IS-IS:200 LSA type: 4 LSA ID:
IS-IS:400, Description: This command displays the ISIS database,
including the LSA type and ID
```

Resolution Steps

Step 1: Step: Verify BGP routing table stability, Command: show bgp summary, Description: Verify that the BGP routing table is stable and not experiencing any churn

Step 2: Step: Verify OSPF adjacency stability, Command: show ospf neighbor, Description: Verify that OSPF adjacencies are stable and not experiencing any loss

Step 3: Step: Verify ISIS LSP stability, Command: show isis database, Description: Verify that ISIS LSPs are stable and not experiencing any flooding

Step 4: Step: Disable BGP route reflection, Command: no bgp router-id 10.10.10.10, Description: Disable BGP route reflection to prevent BGP route churn

Step 5: Step: Reset OSPF adjacency, Command: clear ospf neighbor 10.10.10.10, Description: Reset OSPF adjacency to prevent OSPF adjacency loss

Step 6: Step: Clear ISIS LSPs, Command: clear isis database, Description: Clear ISIS LSPs to prevent ISIS LSP flooding

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 123
Router ID is 10.10.10.10 BGP routing-table version is 123 Router ID
10.10.10.10, RIBs: 2 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ
Up/Down State PrefRcvd, Description: Verify that the BGP routing table
is stable and not experiencing any churn

Command: show ospf neighbor, Expected Output: OSPF neighbor table
Neighbor State Address Interface Pri DeadTime Time Up/Down
---------------------- 10.10.10.10 FULL/DR 10.10.10.10 Eth1/1 1
00:00:03 00:00:01 00:05:30 00:00:02, Description: Verify that OSPF
adjacencies are stable and not experiencing any loss

Command: show isis database, Expected Output: ISIS database LSA type:
1 LSA ID: IS-IS:100 LSA type: 2 LSA ID: IS-IS:200 LSA type: 4 LSA ID:
IS-IS:400, Description: Verify that ISIS LSPs are stable and not
experiencing any flooding
```

Roll-back Procedures

- Step: Restore BGP route reflection, Command: bgp router-id 10.10.10.10, Description: Restore BGP route reflection to prevent BGP route churn

- Step: Reset OSPF adjacency, Command: clear ospf neighbor 10.10.10.10, Description: Reset OSPF adjacency to prevent OSPF adjacency loss

- Step: Rebuild ISIS LSPs, Command: clear isis database, Description: Rebuild ISIS LSPs to prevent ISIS LSP flooding